

Loay Mualem

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Optimization | Machine/Deep learning | Provable data
summarization | Robotics

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Bio

I am a Postdoctoral Researcher at the University of Stuttgart working with Prof. Mathias Niepert. Previously, I completed my Ph.D. in Computer Science at the University of Haifa, where I was supervised by Prof. Moran Feldman at the Computer Science Department and Prof. Dan Feldman at the Robotics & Big Data Labs. My research interests include optimization, machine learning, and large-scale data systems. I focus on developing provable optimization techniques to improve both the efficiency and performance of real-world machine and deep learning models.

Academic Positions

Postdoctoral Researcher, Computer Science, University of Stuttgart, Stuttgart, Germany December'25 – Present
Host: Prof. Mathias Niepert

Education

Ph.D. in Computer Science, University of Haifa, Haifa, Israel October'21 – October'25
Thesis: Subset Selection Techniques for Deep/Machine Learning Applications: From Theory to Practice

M.Sc. in Computer Science, University of Haifa, Haifa, Israel (Summa Cum Laude, Grade: 96/100) March'18 – March'21
Thesis: Online Packing of Squares and Cubes

B.Sc. in Computer Science, University of Haifa, Haifa, Israel (Cum Laude, Grade: 91/100) October'14 – March'18

Relevant Experience

Researcher April'23 - August'25
DataHeroes

- Implemented subset selection techniques for accelerating the training process of deep computer vision models while maintaining a small drop in accuracy.
- Exploring optimization techniques for accelerating the fine-tuning of Large Language Models, with a focus on maintaining model accuracy and robustness.

August'22

EPFL - Swiss Federal Institute of Technology Lausanne
Bernoulli Center Summer School on Modern Trends in Combinatorial Optimization

Teaching (Computer Science Department, University of Haifa)

- 2022-2025: Introduction to Computer Science - Lecturer.
- 2021-2023: Introduction to Mathematics for first-year students - Lecturer.
- 2021-2024: B.Sc. final projects supervisor and mentor.
- 2017-2021: Algorithm Design and Analysis - Teaching Assistant.
- 2016-2017, 2021-2022: Data Structures - Teaching Assistant.
- 2018-2019: Introduction to Computer Science - Teaching Assistant.
- 2019-2020: Automata theory - Teaching Assistant.
- 2016-2017: Introduction to Compilers - Teaching Assistant.

Publications (* stands for equal contribution)

- [1] L. Epstein and L. Muelem. “Online bin packing of squares and cubes”. In: *Algorithmica* (2022), pp. 1–44.
- [2] L. Muelem and M. Feldman. “Using partial monotonicity in submodular maximization”. In: *Advances in Neural Information Processing Systems* 35 (2022), pp. 2723–2736.
- [3] M. Tukan*, L. Muelem*, and A. Maalouf*. “Pruning neural networks via coresets and convex geometry: Towards no assumptions”. In: *Advances in Neural Information Processing Systems* 35 (2022), pp. 38003–38019.
- [4] L. Muelem and M. Feldman. “Resolving the approximability of offline and online non-monotone dr-submodular maximization over general convex sets”. In: *International Conference on Artificial Intelligence and Statistics*. PMLR. 2023, pp. 2542–2564.
- [5] L. Muelem, E. R. Elenberg, M. Feldman, and A. Karbasi. “Submodular Minimax Optimization: Finding Effective Sets”. In: *International Conference on Artificial Intelligence and Statistics*. PMLR. 2024, pp. –.
- [6] M. Goldenberg, L. Muelem, A. Shahar, S. Snir, and A. Akavia. “Privacy-preserving biological age prediction over federated human methylation data using fully homomorphic encryption”. In: *Genome Research* 34.9 (2024), pp. 1324–1333.
- [7] L. Muelem, M. Tukan, and M. Feldman. “Bridging the Gap between General and Down-Closed Convex Sets in Submodular Maximization”. In: *Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, IJCAI-24*. Ed. by K. Larson. Main Track. International Joint Conferences on Artificial Intelligence Organization, Aug. 2024, pp. 1926–1934. DOI: [10.24963/ijcai.2024/213](https://doi.org/10.24963/ijcai.2024/213). URL: <https://doi.org/10.24963/ijcai.2024/213>.
- [8] R. Muelem, L. Morales-Quezada, R. H. Farraj, S. Shance, D. H. Bernshtein, S. Cohen, L. Muelem, N. Salem, R. R. Yehuda, Y. Zbedat, et al. “Econeurobiology and brain development in children: key factors affecting development, behavioral outcomes, and school interventions”. In: *Frontiers in Public Health* 12 (2024), p. 1376075.
- [9] M. Tukan, L. Muelem, and M. Feldman. “Practical 0.385-Approximation for Submodular Maximization Subject to a Cardinality Constraint”. In: *NeurIPS* (2024). In Press.

Submitted

- [1] G. Abo Ayoub, M. Tukan, and L. Muelem. *SpurAudio: A Benchmark for Studying Shortcut Learning in Few-Shot Audio Classification*.
- [2] M. Tukan, F. Fares, Y. Grufinkle, I. Talmor, L. Muelem, V. Braverman, and D. Feldman. *ORBSLAM3-Enhanced Autonomous Toy Drones: Pioneering Indoor Exploration*.

Honors & Fellowships

- **Council for Higher Education (CHE) Fellowship for Outstanding Postdoctoral Researchers in AI and Data Science** (2025). Awarded annually to up to six postdoctoral researchers in Israel working in AI and Data Science.
- **Fulbright Postdoctoral Fellowship** (Opted out). Up to sixteen awards granted annually in Israel.
- **Council for Higher Education (CHE) Fellowship for Outstanding Postdoctoral Researchers from Diverse Populations** (Opted out). Awarded annually to up to eleven postdoctoral researchers from diverse populations in Israel.
- **Mehrez Prize** (2022). Awarded by the Operations Research Society of Israel (ORSIS) for the best paper by a graduate student.
- **NeurIPS Scholar Award** (2022).
- **Uri Peled Memorial Prize** (2022). Awarded by the Caesarea Rothschild Institute for interdisciplinary research in Computer Science and Mathematics.
- **Excellence Scholarship for M.Sc. Candidates** (2017).
- **Council for Higher Education Scholarship for Outstanding Arab M.Sc. Students** (2018).

Research Fields

- Accelerating Deep and Generative Models.
- Autonomous Navigation and Robotics.
- Submodular Functions.
- Eco-Neurobiology and Brain Development.

Volunteering activity

NeurIPS Outreach Program

November'22

Broaden participation in machine learning and artificial intelligence by engaging with high-school students.

Review experience

- SIAM Symposium on Simplicity in Algorithms (SOSA) 2021
- Conference on Neural Information Processing Systems (NeurIPS) 2021,2023,2024
- International Conference on Machine Learning (ICML) 2022,2023
- International Colloquium on Automata, Languages and Programming (ICALP) 2022M
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2023
- The International Conference on Learning Representations (ICLR) 2024
- International Conference on Robotics and Automation (ICRA) 2024,2025
- SIAM Journal on Discrete Mathematics (SIDMA) 2024
- Association for the Advancement of Artificial Intelligence 2025

Skills & Interests

Software:	Python, Julia, Numpy, Scipy, C/C++, Matlab, PyTorch, Java, CVXPy, CVX, SQL, MySQL.
Courses:	To name a few: Introduction to machine learning, Image processing, Introduction to AI, Big-data management, Discrete mathematics, Advanced data structures, Advanced robotics Lab.
Languages:	Arabic (native), Hebrew (fluent), English (fluent)
Interests:	Travelling, sports (swimming, running, tennis, football, basketball), bodybuilding, riddle solving.